

A Multimodal Dataset for Visually Grounded Ambiguity in Machine Translation

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Dataset: <https://huggingface.co/datasets/p1k0/visually-dependent-ambiguity>

Abstract

Ambiguity resolution is a key challenge in multimodal machine translation (MMT), where models must genuinely leverage visual input to map an ambiguous expression to its intended meaning. Although prior work has proposed disambiguation-oriented benchmarks that provide supportive evidence for the role of vision, we observe substantial issues in data quality and a mismatch with translation scenarios. Moreover, existing ambiguity-oriented evaluations are not well suited to broader ambiguity types in open-ended translation. To address these limitations, we present **VIDA** (Visually-Dependent Ambiguity), a dataset of 2,500 carefully curated instances in which resolving an annotated ambiguous source span requires visual evidence. We further propose **Disambiguation-Centric Metrics** that use an LLM-as-a-judge classifier to verify whether annotated ambiguous expressions are resolved correctly at the span level. Experiments with two state-of-the-art Large Vision Language Models under vanilla inference, supervised fine-tuning (SFT), and our chain-of-thought SFT (CoT-SFT) show that while SFT improves overall translation quality, CoT-SFT yields more consistent gains in disambiguation accuracy, especially on out-of-distribution subsets, indicating a stronger generalization for resolving diverse ambiguity types.

1 Introduction

Multimodal machine translation (MMT) extends neural machine translation by incorporating visual context to improve translation quality (Lala and Specia, 2018; Yao and Wan, 2020). Recent Large Vision Language Models (LVLMs) show impressive performance on MMT benchmarks (Bai et al., 2025; Zhu et al., 2025). However, it is still unclear whether LVLMs truly leverage visual information during translation. For instance, prior studies (Elliott, 2018; Wu et al., 2021) show that replacing or perturbing images often leads to only minor degra-

dations, raising questions about the actual contribution of the visual modality.

To investigate this question, recent works (Ma et al., 2024; Wang et al., 2024a) have adopted ambiguity resolution as a probe of visual dependence in MMT, introducing benchmarks such as 3AM (Ma et al., 2024) and MMA (Wang et al., 2024a). 3AM targets English–Chinese translation and primarily focuses on word-level ambiguity, but contains many instances that remain resolvable from text alone. MMA focuses on sentence-level ambiguity in a VQA-style format, which is not directly aligned with the translation scenario. Additionally, prior work on MMT ambiguity evaluation spans several paradigms, including contrastive evaluation with predefined translation candidates (Futeral et al., 2023), as well as task-specific metrics based on predefined variants or rule-based matching for narrower ambiguity phenomena (Lala and Specia, 2018; Li et al., 2021). While these settings provide ambiguity-oriented evaluation, they are less suitable for open-ended MMT generation with broader ambiguity types and valid paraphrases. General MT metrics such as BLEU and COMET are also useful for overall translation quality, but they are not designed to measure whether a particular ambiguous span has been resolved correctly.

To reliably evaluate visual disambiguation in MMT, we need both an MMT dataset with visually-dependent ambiguities and an evaluation method that directly measures disambiguation accuracy. In this work, we address these limitations by introducing **VIDA** (Visually-Dependent Ambiguity), a dataset of 2,500 instances curated to capture visually-dependent translation ambiguities. VIDA comprises three subsets: VIDA-Base, VIDA-Sent, and VIDA-CollN, covering word- and sentence-level ambiguities, as well as collective-noun cases where visual context is necessary to translate abstract entities into concrete target expressions. We further propose **Disambiguation-Centric Metrics**

based on an LLM-as-a-judge classifier. Unlike standard translation metrics, our metrics explicitly target disambiguation accuracy and provide a more sensitive measure of correct ambiguity resolution.

Beyond dataset and evaluation, we further explore a chain-of-thought supervised fine-tuning (CoT-SFT) strategy (Muennighoff et al., 2025) to examine whether reasoning supervision can elicit visually grounded disambiguation in MMT. Specifically, we manually synthesize task-specific reasoning traces that guide the model to identify the ambiguous expression and justify its resolution with visual evidence before generating the final translation. Experiments show that CoT-SFT yields consistent gains in disambiguation accuracy over standard SFT, suggesting better generalization across diverse ambiguity types. The main contributions of this paper are as follows.

- We introduce **VIDA**, an MMT dataset featuring visually-dependent ambiguities at both word and sentence levels.
- We propose **Disambiguation-Centric Metrics** to directly measure disambiguation accuracy, complementing standard translation metrics for MMT disambiguation.
- We further explore a CoT-SFT method by augmenting training with synthetic reasoning traces for MMT disambiguation.

2 Related Works

Multimodal Disambiguation Datasets. Recent benchmarks such as 3AM (Ma et al., 2024) and MMA (Wang et al., 2024a) address multimodal disambiguation but show limitations in MMT evaluation. 3AM contains instances that are resolvable by text alone, reducing visual sensitivity to visual grounding, while MMA uses a VQA-style format incompatible with translation scenarios. These limitations motivate our **VIDA** dataset, which covers both word- and sentence-level cases strictly requiring visual evidence for correct interpretation.

Evaluation Metrics for MMT. Prior work on multimodal ambiguity evaluation has used several different settings. CoMMuTE (Futeral et al., 2023) evaluates disambiguation with predefined translation candidates rather than open-ended translation generation, while MLT (Lala and Specia, 2018) and AmbigCaps (Li et al., 2021) evaluation focus on word-level ambiguity and rely on

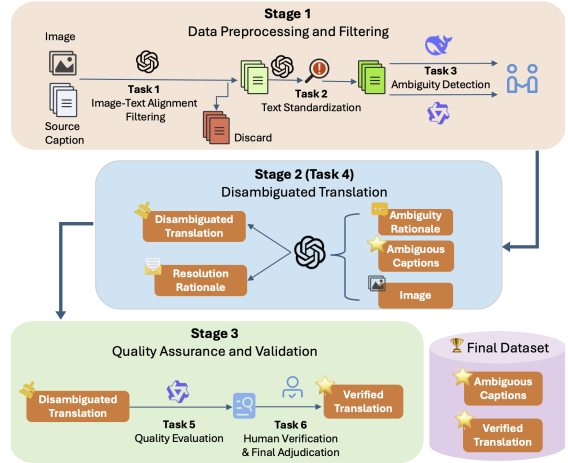


Figure 1: Three-stage VIDA curation pipeline

rule-based string matching. Furthermore, standard MT metrics such as BLEU (Papineni et al., 2002) and COMET (Rei et al., 2020) do not directly verify whether an ambiguous span has been resolved correctly, since surface-overlap metrics may penalize valid paraphrases or lexical variation and sentence-level metrics are too coarse-grained for span-level disambiguation. In this work, we propose Disambiguation-Centric Metrics using an LLM-as-a-judge classifier to directly assess span-level disambiguation accuracy in open-ended MMT generation.

3 Dataset Curation

To construct MMT dataset with visually-dependent translation ambiguities, we adopt a three-stage semi-automatic pipeline, as shown in Figure 1. Our goal is to collect instances where the disambiguating evidence is grounded in the image, and to provide human-verified reference translations.

3.1 Stage 1: Data Preprocessing and Filtering

The goal of this stage is to curate image-text aligned and textually clean source captions with visually-dependent ambiguities. We use GPT-4o (Hurst et al., 2024) to filter mismatched image-text pairs and normalize text. We then apply a dual-model consensus strategy with two state-of-the-art commercial LLMs, Qwen-Max (Team, 2024) and DeepSeek-v3 (Liu et al., 2024): each model independently assesses whether the caption remains ambiguous without the image. We retain only captions that both models judge as ambiguous, and record an *Ambiguous Caption* and an *Ambiguity Rationale* that points the ambiguous span and the

disambiguation cue.

3.2 Stage 2: Disambiguated Translation

In this stage, we produce high-quality disambiguated translations for each retained caption. For each caption from Stage 1, we generate a disambiguated translation using GPT-4o with a structured input that includes the *Ambiguous Caption*, the paired *Image*, and the *Ambiguity Rationale* from the previous stage. The model outputs both a *Disambiguated Translation* that resolves the ambiguity and a *Resolution Rationale* explaining how visual information is used to support the decision.

3.3 Stage 3: Quality Assurance and Validation

This stage finalizes the dataset with an LLM-as-a-judge quality check followed by human verification, ensuring both translation quality and correct visual disambiguation. We employ Qwen-Max to evaluate each disambiguated translation along two dimensions: semantic preservation and fluency. For each dimension, Qwen-Max outputs a score from 1 to 5 together with a brief justification. We flag cases with scores below 4 as potentially problematic and prioritize them for closer inspection.

To ensure translation correctness, we further conduct human verification with two native Chinese speakers (one Ph.D. and one M.S. in Computer Science). For each instance, annotators are shown the image, the source caption, the ambiguity rationale, the candidate translation, and Qwen-Max’s judgment. Each annotator independently reviews the translation using three criteria: (i) whether the annotated ambiguity is correctly resolved, (ii) whether the translation is fluent, and (iii) whether the meaning of the translation is preserved. If the candidate fails any criterion, the annotator provides a corrected translation based on the image and the ambiguity rationale. During verification, we identified collective noun cases where underspecified terms must be concretized based on the image content and group these cases into the **Collective Noun Subset**. Finally, **VIDA** comprises 2,500 instances across three subsets. See Appendix A for details.

4 Evaluation Metrics for Disambiguation

In MMT disambiguation, the objective of evaluation is to measure disambiguation accuracy rather than overall translation quality. We use an LLM-as-a-judge to classify whether the ambiguous span is correctly resolved, and compute disambiguation

accuracy from the binary outputs. Specifically, we fine-tune Qwen3-8B on **VIDA** to determine whether annotated ambiguous expressions are correctly resolved in translation. We train the classifier in a contrastive setting. Gold disambiguated translations are used as positive samples. Negative samples are candidate translations from our curation pipeline that fails to resolve the annotated ambiguity. Building on this, we introduce two complementary **Disambiguation-Centric Metrics** for a comprehensive evaluation of disambiguation performance: **Disambi-Term** measures term-level disambiguation accuracy by evaluating each annotated ambiguous term in the entire dataset. **Disambi-Inst.** reports instance-level accuracy, counting a sentence as correct only if all ambiguous expressions within it are correctly resolved.

5 Experiments

5.1 Experimental Settings

Dataset We conduct experiments on the VIDA dataset. VIDA-Base is split into training and test sets with a 7:3 ratio, resulting in 1,352 training samples and 580 test samples. Due to their limited size, VIDA-CollN and VIDA-Sent are used exclusively as out-of-distribution (OOD) test sets.

Metrics We report standard translation metrics to assess overall translation quality, including *BLEU*, *chrF* (Popović, 2015), *chrF++* (Popović, 2017), *METEOR* (Banerjee and Lavie, 2005), *TER* (Snover et al., 2006), *BERT-F1* (Devlin et al., 2019) and *COMET*. More importantly, we evaluate disambiguation performance using the proposed Disambiguation-Centric Metrics, Disambi-Term and Disambi-Inst.

Models and Baselines We evaluate two state-of-the-art LVLMs: **Qwen2.5-VL-7B** (Bai et al., 2025), **InternVL3-8B** (Zhu et al., 2025). For each model, we consider three settings: (i) the vanilla model without task-specific fine-tuning, (ii) supervised fine-tuning (SFT), and (iii) the Chain-of-Thought Supervised Fine-Tuning (CoT-SFT), which augments SFT with manually synthesized reasoning traces for ambiguity resolution (Appendix C).

5.2 Experimental Results

Table 1 summarizes results for two LVLMs under Vanilla, SFT, and CoT-SFT on in- and out-of-distribution tests.

Model	Dataset	Model Setting	BLEU	chrF	chrF++	TER	BERT-F1	METEOR	COMET	Disambi-Term	Disambi-Inst.
InternVL3-8B	All-Test	Vanilla	48.04	41.95	32.98	40.29	86.63	58.47	84.49	50.86	39.81
		SFT	49.20	42.73	32.82	43.44	87.07	58.95	85.55	54.36	43.77
		CoT-SFT	47.64	41.16	32.99	41.61	87.18	58.78	85.88	58.45	48.78
	VIDA-Base-Test	Vanilla	53.51	46.76	36.87	35.66	88.84	65.24	86.08	60.18	46.55
		SFT	55.31	48.31	37.13	34.28	89.61	66.55	87.30	62.67	50.17
		CoT-SFT	51.10	44.41	35.75	38.06	88.56	63.25	86.44	64.89	51.38
	VIDA-Sent	Vanilla	42.51	36.85	33.01	44.76	84.31	52.54	84.21	50.00	50.00
		SFT	36.99	35.69	31.96	67.93	83.93	51.67	84.70	55.45	55.45
		CoT-SFT	44.22	38.19	34.28	45.27	85.70	55.32	86.39	58.97	58.97
	VIDA-CollN	Vanilla	36.56	31.66	27.14	49.24	84.63	48.79	81.39	18.36	12.16
		SFT	37.97	32.92	28.22	49.11	85.11	50.56	82.60	22.62	14.90
		CoT-SFT	39.26	33.89	25.70	48.52	85.70	51.05	83.96	38.36	32.55
Qwen2.5-VL-7B	All-Test	Vanilla	47.85	41.67	34.12	42.74	86.56	58.88	84.83	50.08	39.81
		SFT	49.13	42.85	34.58	40.34	87.38	59.51	85.82	52.81	42.46
		CoT-SFT	47.59	41.39	33.26	42.49	87.06	58.60	85.84	55.51	46.08
	VIDA-Base-Test	Vanilla	52.38	45.66	37.66	37.64	88.53	64.45	86.30	58.49	46.38
		SFT	53.88	47.09	38.35	36.17	89.11	65.48	87.07	61.42	49.31
		CoT-SFT	50.41	44.57	36.00	39.44	88.32	62.75	86.35	60.71	46.90
	VIDA-Sent	Vanilla	44.46	38.92	34.97	50.95	84.21	54.78	84.41	51.28	51.28
		SFT	45.12	39.52	35.59	42.66	85.86	55.41	86.06	52.56	52.56
		CoT-SFT	45.54	39.79	35.43	45.17	85.66	55.02	86.41	60.26	60.26
	VIDA-CollN	Vanilla	38.06	32.83	24.63	50.54	84.87	51.16	82.06	19.02	12.16
		SFT	39.02	33.71	25.28	49.24	85.30	50.96	82.69	21.31	14.51
		CoT-SFT	38.21	32.49	24.51	50.71	85.32	51.34	83.39	33.77	27.45

Table 1: Performance comparison of **InternVL3-8B** and **Qwen2.5-VL-7B** under Vanilla, SFT, and CoT-SFT settings. **Box** highlights best standard metrics, **Box** highlights best disambiguation metrics.

Analysis on In-Distribution Dataset. Evaluation on the in-distribution dataset (**VIDA-Base-Test**) serves to examine how well the models adapt to the training distribution (VIDA-Base-Train). On VIDA-Base-Test, SFT setting achieves the strongest overall translation quality for both models across most standard metrics. CoT-SFT remains competitive on semantic metrics (e.g., COMET) for both models, but can slightly underperform SFT on surface-overlap metrics such as BLEU, likely because ambiguity resolution often requires paraphrasing or structural reformulation that deviates from reference wording. Under Disambiguation-Centric evaluation, CoT-SFT yields the best performance for InternVL3-8B (Disambi-Term/Inst. **64.89/51.38**), surpassing both SFT and Vanilla. In contrast, for Qwen2.5-VL-7B, CoT-SFT improves over Vanilla but remains slightly below SFT (60.71/46.90 vs. 61.42/49.31), suggesting that in-distribution gains from reasoning supervision are model-dependent. We attribute part of the CoT-SFT disambiguation gap to *overthinking* behaviors and provide qualitative analysis in Appendix D.

Analysis on Out-of-Distribution Dataset. Evaluation on the out-of-distribution (OOD) datasets (**VIDA-Sent** and **VIDA-CollN**) tests whether models can generalize the disambiguation to unseen ambiguity types. Compared to in-distribution results, CoT-SFT shows clearer advantages under

distribution shift. For standard translation metrics, CoT-SFT achieves the highest COMET on both OOD subsets for both models, indicating stronger semantic adequacy on unseen ambiguity types. Disambiguation-centric metrics make the advantage more explicit. Compared to SFT, CoT-SFT achieves consistent gains on VIDA-Sent (approximately **+3.5** points for InternVL3-8B and **+7.5** points for Qwen2.5-VL-7B on both Disambi-Term and Disambi-Inst.), and the improvements are substantially larger on VIDA-CollN (over **+15** points for InternVL3-8B and over **+12** points for Qwen2.5-VL-7B), suggesting that CoT-SFT demonstrates stronger generalization beyond the training distribution.

Analysis on All-Test Dataset. All-Test merges all subsets into a single evaluation rather than averaging their scores, reflecting overall performance weighted by subset sizes. On standard translation metrics, SFT consistently improves over Vanilla for both models, indicating better overall translation quality on the mixed test distribution. In contrast, CoT-SFT consistently yields higher semantic adequacy and disambiguation than SFT: it achieves the highest COMET and improves Disambi-Term/Inst. by about **+4.1/+5.0** for InternVL3-8B and **+2.7/+3.6** for Qwen2.5-VL-7B. Overall, CoT-SFT achieves a strong balance between semantic adequacy and disambiguation and

transfers better to the diverse ambiguity types.

6 Conclusion

We introduce VIDA, an MMT dataset with visually-dependent translation ambiguities covering both word- and sentence-level, and propose Disambiguation-Centric Metrics that directly measure disambiguation accuracy using an LLM-as-a-judge classifier. We further compare CoT-SFT against SFT for disambiguation. Our findings show that SFT achieves strong overall translation quality on in-distribution data, whereas CoT-SFT achieves larger gains in disambiguation accuracy on the mixed All-Test evaluation, suggesting better generalization for resolving diverse ambiguity types.

Limitations

Although **VIDA** targets highly visually dependent ambiguities, the dataset remains modest in scale compared to large-scale translation corpora and is currently limited to English–Chinese translation. In addition, the reasoning traces used to supervise CoT-SFT are synthetically constructed based on manually designed patterns, which may not reflect natural model reasoning. Future work includes expanding the dataset to additional language pairs and exploring training paradigms that encourage models to develop task-relevant reasoning behaviors without relying on handcrafted traces, such as reinforcement learning or preference-based optimization methods.

Ethics Statement

The proposed **VIDA** dataset is curated from publicly available data sources. The dataset will be released with usage guidelines to support research on multimodal machine translation and ambiguity resolution.

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A Dataset Statistics

A.1 Pipeline Filtering Statistics

The large reduction from the raw candidate pool to the final benchmark reflects the strict quality control in our curation pipeline. We begin with 26,452 raw image–text pairs, retain 14,993 after image–text matching and text normalization, and then keep only 2,500 instances after ambiguity detection and dual-LLM verification. Subsequent stages refine translations and annotations without discarding additional instances. All 2,500 retained instances are then human-verified by two native Chinese annotators, corresponding to 100% human verification coverage. The filtering funnel is summarized in [Table 2](#).

A.2 VIDA Subset Statistics

The rigorous pipeline outlined in [section 3](#) results in the construction of a new dataset, **VIDA** (Visually-Dependent Ambiguity). In total, **VIDA** comprises 2,500 instances specifically curated to feature high ambiguity complexity and visual dependency. The dataset comprehensively covers both word-level and sentence-level ambiguities and is organized into the following three subsets:

- **VIDA-Base:** Curated from the 3AM dataset, this subset contains **1,932 samples**, primarily focusing on word-level ambiguities that require visual context for resolution.
- **VIDA-CollN (Collective Noun Subset):** This specialized subset consists of **256 samples** focusing on the disambiguation of collective nouns, where the abstract nature of the group is made concrete by the associated visual information.

Stage	Remaining	Filtered Out
Initial raw pairs	26,452	–
After image–text filtering	14,993	11,459
After ambiguity verification	2,500	12,493
Human verification coverage	2,500 (100%)	0

Table 2: Filtering funnel for constructing VIDA.

Subset	Ambiguity Focused	Size	Avg. Length (words)	Avg. Ambi. Terms
VIDA-Base	Word-Level	1,932	11.12	1.7826
VIDA-Sent	Sentence-Level	312	6.00	1.00
VIDA-CollN	Word-Level	256	10.08	1.20

Table 3: Statistical summary of **VIDA** subsets

- **VIDA-Sent:** Adapted from the MMA dataset, this subset provides **312 samples**. These instances tend to exhibit more complex, sentence-level semantic ambiguities that necessitate a holistic understanding of the image for correct interpretation and translation.

A complete statistical summary of these subsets is provided in Table 3. For each subset, the column *Ambiguity* specifies the primary ambiguity focus (word-level, sentence-level, or mixed). *Size* denotes the total number of samples in each subset. *Avg. Length (Words)* gives the average sentence length measured by the number of words, and *Avg. Ambi.* indicates the average number of ambiguities per sentence.

VIDA-Base is the largest subset (1,932 samples), consisting primarily of word-level ambiguities. It contains relatively longer sentences, averaging 11.12 words, and exhibits the highest ambiguity density (1.78 ambiguous terms per sentence). **VIDA-Sent** (312 samples) specifically focuses on sentence-level ambiguities, with shorter sentences averaging 6.00 words and exactly one annotated ambiguity per instance. Finally, **VIDA-CollN** (256 samples) also targets word-level ambiguities, specializing in collective nouns. Compared to VIDA-Base, VIDA-CollN features shorter sentences (10.08 words on average) and a lower ambiguity density (1.20 per sentence).

B Additional Validation and Judge Details

B.1 Visual Dependence Ablations

To further verify that VIDA instances require the correct visual context, we evaluate models under two ablated settings on **All-Test**: (i) **Text-only**, where the paired image is removed and replaced by

the corresponding text-only backbone, and (ii) **Random Image**, where the correct image is replaced with a randomly sampled image from the test set. Results are shown in Table 4 and Table 5. In both cases, performance drops consistently on both standard MT metrics and the proposed disambiguation-centric metrics. These degradations clearly demonstrate the visual dependence of VIDA: the ambiguous text in our dataset must rely on the correct image information to be successfully disambiguated, and performance drops markedly when the model is given no image or a random image.

B.2 Judge Reliability

We further validate the reliability of the Judge from two complementary perspectives. First, we perform **Cross-Model Judge Verification** by re-evaluating the exact same translation outputs from Qwen2.5-VL-7B (CoT-SFT) on **All-Test** with a second Judge based on LLaMA3.1-8B. As shown in Table 6, the scores remain highly consistent across judge backbones. Second, we measure **Human-Judge Alignment** on a randomly sampled 20% subset of VIDA (500 instances) and compute Cohen’s Kappa between expert human judgments and the Qwen-based Judge. Table 7 shows strong agreement across all subsets, supporting that the Judge captures semantic disambiguation resolution rather than model-specific preferences.

B.3 Judge Training Details

The Judge is fine-tuned from Qwen3-8B as a binary classifier for span-level disambiguation verification. Each training instance contains the source caption, a candidate translation, the annotated ambiguous span, and the gold interpretation of that span; the model predicts whether the ambiguity is correctly resolved in the translation. Gold disambiguated translations are used as positive examples. Negative examples are drawn from candidate translations produced during the curation pipeline that fail to resolve the annotated ambiguity. To reduce false negatives, we further filter out semantically equivalent candidates with an auxiliary LLM-based screening step before training.

We fine-tune the Judge for 10 epochs with learning rate 1×10^{-5} and batch size 8. LoRA is used with rank 8, alpha 32, and dropout 0.1. The final training set is constructed with a positive-to-negative ratio of 2:1. This contrastive setup encourages the Judge to focus on disambiguation correctness rather than superficial lexical overlap, which

Model	BLEU	chrF	chrF++	TER	BERT-F1	METEOR	COMET	Disambi-Term	Disambi-Inst.
InternVL3-8B vs. Qwen2.5-7B									
LVLm	48.04	41.95	32.98	40.29	86.63	58.47	84.49	50.86	39.81
↑ vision ↑	6.88	6.48	2.26	-5.83	2.03	6.63	2.51	8.03	8.54
LLM	41.16	35.47	30.72	46.12	84.60	51.84	81.98	42.83	31.27
Qwen2.5-VL-7B vs. Qwen2.5-7B									
LVLm	47.85	41.67	34.12	42.74	86.56	58.88	84.83	50.08	39.81
↑ vision ↑	4.27	3.59	0.89	-1.76	1.76	5.97	1.73	7.25	12.18
LLM	41.16	35.47	30.72	46.12	84.60	51.84	81.98	42.83	31.27

Table 4: Ablation results without images on **All-Test**. The middle row reports the gain from adding the correct visual input over the text-only baseline.

Model	Image Setting	BLEU	chrF	chrF++	TER	BERT-F1	METEOR	COMET	Disambi-Term	Disambi-Inst.
InternVL3-8B	Correct image	48.04	41.95	32.98	40.29	86.63	58.47	84.49	50.86	39.81
	Random image	44.51	38.79	32.12	41.89	85.53	54.67	82.49	42.39	30.51
Qwen2.5-VL-7B	Correct image	47.85	41.67	34.12	42.74	86.56	58.88	84.83	50.08	39.81
	Random image	43.18	37.71	32.30	43.82	85.18	53.83	82.26	42.81	32.08

Table 5: Ablation results with random images on **All-Test**. Replacing the paired image with a random one substantially degrades both standard MT metrics and disambiguation-centric metrics.

Judge	Disambi-Term	Disambi-Inst.
Qwen-Judge	55.51	46.08
LLaMA-Judge	55.72	47.69

Table 6: Cross-model judge verification on the same Qwen2.5-VL-7B (CoT-SFT) predictions.

Dataset	Cohen’s Kappa
VIDA-Base-Test	0.8828
VIDA-Sent	0.8932
VIDA-CollN	0.9023
Average	0.8928

Table 7: Human-Judge alignment measured by Cohen’s Kappa on a 500-instance sample.

is crucial for handling valid paraphrases and lexical variation in open-ended translation.

C Chain-of-Thought Supervised Fine-Tuning

We define a task-specific six-step reasoning template that guides models to articulate the alignment between ambiguous expressions and visual evidence. Each synthetic trace is constructed according to the following standardized reasoning template:

1. **Visual Grounding:** Examine the image carefully and identify the visual elements that cor-

respond to key words or phrases in the source sentence. Describe how these elements connect to the text.

2. **Initial Translation:** Generate a preliminary translation based on both the text and the grounded visual evidence.
3. **Ambiguity Check:** Review the initial translation and highlight any terms that remain ambiguous—those whose meanings are unclear or context-dependent when relying on text alone.
4. **Visual Disambiguation:** This step is critical. While visual grounding establishes a mapping between the image and the text, the initial translation can still leave some ambiguities unresolved. The model explicitly revisits the image, not only to strengthen the connection between ambiguous terms and their corresponding visual evidence, but also to refresh its access to visual information while mitigating the risk of visual token attention decay during long-sequence generation (Xing et al., 2024; Chu et al., 2025) and hallucination (Wang et al., 2024b). Through this re-examination, the model is better guided to ground its disambiguation decisions in the most relevant visual cues.
5. **Localized Refinement:** Update only the am-

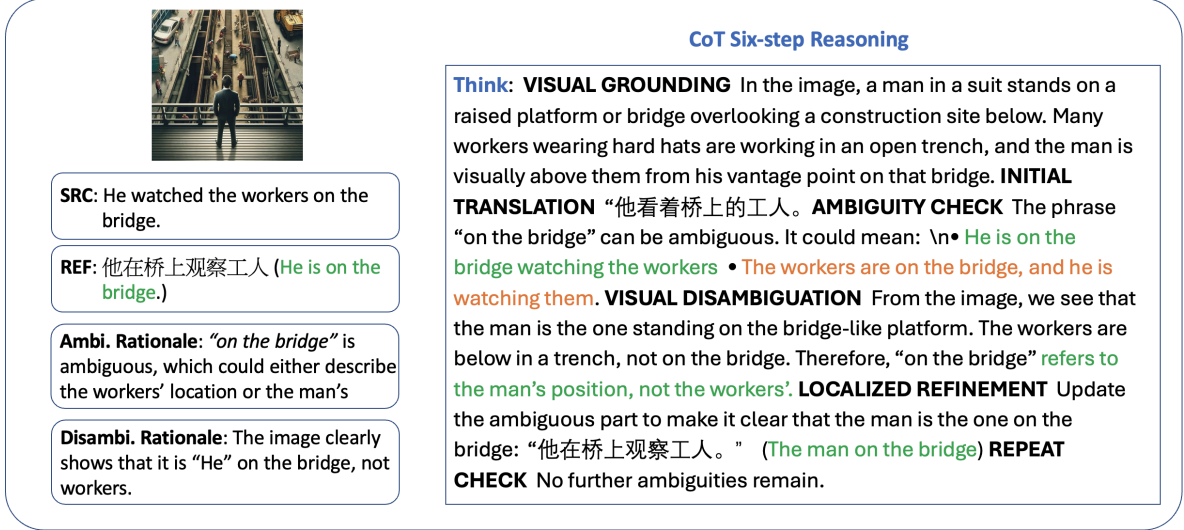


Figure 2: Example of CoT six-step reasoning resolving the ambiguity.

biguous parts of the initial translation while keeping the rest unchanged. This constraint prevents unnecessary modifications to the sentence structure and helps maintain overall translation fluency.

6. **Repeat Check:** Reassess the updated translation. If ambiguities remain, iterate steps 3–5 until the translation is fully disambiguated.

An example is provided in Figure 2.

D Qualitative Analysis

As discussed in section 5, CoT-SFT exhibits a strong ability to enhance disambiguation performance, particularly on challenging OOD subsets (**VIDA-Sent**, **VIDA-CollN**). This raises a key question: *how does explicit reliance on visual information shape the model’s reasoning?* Figure 3 (left and middle) illustrates two case studies that shed light on this process, showing how CoT-SFT aligns ambiguous terms with visual evidence in **VIDA-CollN** and **VIDA-Sent**.

The **VIDA-CollN** example (left of Figure 3) illustrates the collective noun ambiguity, which the source sentence contains the ambiguous noun “object”, which requires a concrete translation (“paddle”). The SFT model, without reasoning, outputs the literal “object,” which fails to capture the intended meaning. In contrast, the CoT-SFT shows that the model first generates an initial translation (物体), maintaining the literal meaning. During the ambiguity check, the model detects that “object” is

ambiguous. In the subsequent visual disambiguation step, it grounds the word to the image and identifies that the woman is holding a paddle. Finally, in localized refinement, the model updates the translation to “paddle”, producing the correct disambiguated output.

The **VIDA-Sent** example (middle of Figure 3) demonstrates sentence-level ambiguity where an idiomatic expression could be misunderstood literally. The phrase “got a green thumb” could be interpreted literally or idiomatically. The SFT model again produces a literal output of the color of thumb in Chinese. In contrast, the CoT-SFT first provides a literal initial translation (绿色的手). Through visual disambiguation, it recognizes from the image that the woman is gardening, and therefore refines the output to “Gardening expert”, correctly capturing the idiomatic meaning.

Although CoT-SFT can improve disambiguation accuracy, we observe that it may also introduce *overthinking* on relatively straightforward inputs, which can degrade translation quality. In our reasoning template, this behavior often arises after the model produces an adequate initial translation: subsequent reasoning steps may overwrite the initial output by injecting unnecessary or spurious reasoning, e.g., overusing irrelevant visual details or interpreting idiomatic expressions too literally, thereby leading to flawed revisions.

The right panel of Figure 3 illustrates an overthinking case. The phrase “iPod’s touch” should be interpreted as “iPod-like touch screen”. The model first provides a reasonable image descrip-



Figure 3: Case study of CoT-SFT vs. SFT

tion and recognizes the intended interpretation during ambiguity checking. However, in the later disambiguation step, it over-interprets the phrase by incorrectly linking it to “*someone physically touching*” mentioned in the grounding step, rather than the relevant cue about the product feature. As a result, the model revises an initially adequate interpretation into an incorrect final translation. This example suggests that excessive reasoning can override correct early hypotheses and partially explains the performance drop observed for CoT-SFT on in-distribution data.